

AP CALCULUS AB – UNIT 3

3.2 Implicit Differentiation



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Differentiate relations in which y is defined implicitly as a function of x .
- Solve algebraically for dy/dx and evaluate it at points on an implicit curve.
- Determine tangent and normal lines, horizontal and vertical tangents, and second derivatives.
- Recognize when a derivative formula is inconclusive at a singular point.

Keys: Essential Knowledge and Vocabulary

Implicit relation

An equation involving both x and y without first solving explicitly for y .

Derivative of a y -term

Apply the Chain Rule: $\frac{d}{dx}(y^n) = ny^{n-1}y'$ and $\frac{d}{dx}(\sin y) = \cos y y'$.

Product containing y

$$\frac{d}{dx}(xy) = xy' + y.$$

Horizontal tangent

Usually numerator = 0 and denominator $\neq 0$ in a simplified formula for y' .

Vertical tangent

Usually denominator = 0 and numerator $\neq 0$.

Second derivative

Differentiate the first-derivative relation again and substitute known values of y' .

Explanation, Strategy, and Common Errors

- Differentiate every term with respect to x and attach y' whenever the differentiated expression contains y .
- Collect all terms containing y' on one side before factoring and dividing.
- Verify that a stated point lies on the original curve before using it.
- For a normal line, use the negative reciprocal of a finite nonzero tangent slope.
- If both numerator and denominator of the derivative formula vanish, return to the original relation; the formula is inconclusive.

Solved Example: Circle and tangent line

For $x^2 + y^2 = 25$, find y' and the tangent line at $(3, 4)$.

Answer and Reasoning

$$2x + 2yy' = 0, \text{ so } y' = -x/y. \text{ At } (3, 4), m = -3/4 \text{ and } y - 4 = -\frac{3}{4}(x - 3).$$

Solved Example: Product and trigonometric y -terms

For $x^2y + \sin y = 4$, find y' .

Answer and Reasoning

$$2xy + x^2y' + \cos y \, y' = 0, \text{ so } y' = -\frac{2xy}{x^2 + \cos y}.$$

Solved Example: Second derivative

For $x^2 + y^2 = 25$, find y'' at $(3, 4)$.

Answer and Reasoning

Differentiate again: $2 + 2(y')^2 + 2yy'' = 0$. With $y' = -3/4$, $y'' = -25/64$.

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. For $x + y = 7$, find y' .

2. For $x^2 + xy + y^2 = 7$, find y' .

3. For $\sin x + \cos y = 1$, find y' .

4. Find the slope of $x^2 + y^2 = 13$ at $(2, 3)$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Write the tangent line to $x^2 + xy = 6$ at $(2, 1)$.

6. For $e^{xy} = x + y$, solve for y' .

7. Find the horizontal tangents on $x^2 + 4y^2 = 16$.

8. Find the vertical tangents on $x^2 + 4y^2 = 16$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. For $x^3 + y^3 = 6xy$, find y' .

10. Find the normal line to $x^2 - xy + y^2 = 3$ at $(1, -1)$.

11. For $x^2 + y^2 = 10$, find y'' at $(1, 3)$.

12. On $x^2 + xy + y^2 = 7$, is the tangent at $(1, 2)$ increasing or decreasing?

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Show that no value of c makes $x^2 + cxy + y^2 = 6$ pass through $(1, 1)$ with slope -2 .

14. For $\ln(x + y) = xy$, find y' .

15. Find the horizontal-tangent points on $x^2 + xy + y^2 = 3$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. At $(0, 0)$ on $y^2 = x^2(x + 1)$, use leading terms to identify candidate tangent slopes.

17. Prove $x^2 + xy + y^2 = 7$ has no point satisfying both horizontal- and vertical-tangent conditions.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. If $x^2 + y^2 = 9$, then y' equals

- | | |
|------------|------------|
| (A) $-x/y$ | (B) $-y/x$ |
| (C) x/y | (D) y/x |

2. The derivative of $\sin y$ with respect to x is

- | | |
|---------------------|--------------------|
| (A) $\cos y$ | (B) $\cos y \, y'$ |
| (C) $-\sin y \, y'$ | (D) $\sin y \, y'$ |

3. For $x^2 + 4y^2 = 16$, a vertical tangent occurs at

- | | |
|--------------|---------------------|
| (A) $(0, 2)$ | (B) $(0, -2)$ |
| (C) $(4, 0)$ | (D) $(2, \sqrt{3})$ |

4. If both numerator and denominator of an implicit derivative formula are zero at a point, then

- | | |
|------------------------------------|----------------------------------|
| (A) the tangent must be horizontal | (B) the tangent must be vertical |
| (C) the point is invalid | (D) the formula is inconclusive |

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $x^2 + y^3 = 6$, at $x = 1$ and $y > 0$, the slope is closest to

- | | |
|-------------|-------------|
| (A) -0.23 | (B) -0.39 |
| (C) -0.57 | (D) -0.82 |

6. On $x^2 + e^y = 4x$, at $(1, \ln 3)$ the second derivative is

- | | |
|-------------|------------|
| (A) $-10/9$ | (B) $-2/3$ |
| (C) $2/3$ | (D) $10/9$ |

AP-Style Free-Response Practice

Calculator Permitted

The curve is $x^2 + y^3 = 6$. Consider the point where $x = 1$ and $y > 0$.

(a) Find the point to three decimals. [2 points]

(b) Find and evaluate dy/dx . [3 points]

(c) Write the tangent-line approximation near $x = 1$. [2 points]

No Calculator

Let C be the curve $x^2 + xy + y^2 = 7$.

(a) Find dy/dx . [2 points]

(b) Find the tangent line at $(1, 2)$. [2 points]

(c) Find all horizontal-tangent points. [3 points]

(d) Explain possible failure where $x + 2y = 0$. [2 points]

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